Timothy Stephens
Curriculum vitae

Department of Biochemistry and Microbiology, Rutgers University, USA

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Personal summary

I am a computational biologist and research leader developing an independent program at the intersection of genomics, evolution, and data science. My work focuses on understanding the origin and function of genes with unknown function (i.e., "dark genes") and building computational approaches that can be applied across diverse biological systems, including microbial, environmental, biomedical, and translational research. I have led collaborative research programs, mentored graduate and undergraduate researchers, and guided projects from scientific question through analysis and publication. I am also a co-founder and former CTO of OceanOmics, where I helped translate research into scalable environmental health technologies and advance multiple platforms to proof of concept.

Education and Research Experience

2024—2026	Research Associate, Department of Biochemistry and Microbiology, Rutgers University
2019—2024	Post Doctoral Associate, Department of Biochemistry and Microbiology, Rutgers University
2016—2019	PhD, Institute for Molecular Bioscience, The University of Queensland
2015	Honours, Institute for Molecular Bioscience, The University of Queensland
2012—2015	Bachelor of Biotechnology, Institute for Molecular Bioscience, The University of Queensland

Selected Grants

2026 New Jersey CSIT Clean Tech R&D Voucher Pilot Program, $39,760 USD
Funds awarded to our startup OceanOmics to utilize government facilities to aid development of our shellfish health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University), New Jersey Commission on Science, Innovation and Technology

2024 New Jersey CSIT Clean Tech R&D Voucher Pilot Program, $39,860 USD
Funds awarded to our startup OceanOmics to utilize government facilities to aid development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University), New Jersey Commission on Science, Innovation and Technology

2024 New Jersey CSIT Clean Tech Seed Grant Program, $74,870 USD
Funds awarded to our startup OceanOmics to continue development of our coral health monitoring toolkit.
Principal Investigators: **Timothy Stephens** (Rutgers University), New Jersey Commission on Science, Innovation and Technology

2023 Rutgers TechAdvance Fund, $74,842 USD
Funds to continue development of our coral health monitoring toolkit.
Principal Investigators: Debashish Bhattacharya (Rutgers University), Rutgers University

Selected Teaching

2020—2026 **Co-lecturer**
Rutgers University
Design and conducted lectures on de novo next-generation genome and transcriptome sequencing, metabolomics, proteomics, metagenomics, phylogenetics, and phylogenomics in a join undergraduate and postgraduate course titled "Fundamentals of Microbial Genomics". Duties include designing and marking student' exam questions and oral presentations.

Selected Talks

2024 **Stephens T. G.**, Kulczyk A. W., and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *3rd Joint Congress on Evolutionary Biology*. 26-30 July 2024, Montreal, Canada.

2023 **Stephens T. G.**, Etten J. V., Benites L. F., Mcdermott T., and Bhattacharya D. Cyanidiophyceae: The extremophilic red algae that underpin hot spring microbial communities in Yellowstone National Park. *UQ Marine Ecogenomics Symposium 2023*. 13-14 December 2023, University of Queensland, Australia.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *International Society of Endocytobiology conference 2023*. 10-14th September 2023, Field Museum, Chicago.

2023 **Stephens T. G.**, Chille E., Strand E. L., Putnam H. M., and Bhattacharya D. Multi-omics investigation of coral resilience. *Invited talk at Carnegie Plant Biology Seminar, August 2023*. 25th August 2023, Carnegie Department of Plant Biology, Stanford University, USA.

2023 **Stephens T. G.**, Calatrava V., Gabr A., Grossman A., and Bhattacharya D. Exploring the origin and evolution of primary plastids using *Paulinella* as a model system. *New Phytologist Next Generation Scientists Conference 2023*. 3rd July 2023, National University of Singapore, Singapore.

2022 **Stephens T. G.**, Strand E. L., Putnam H. M., and Bhattacharya D. Differences in ploidy and the prevalence of clonal propagation between *Montipora capitata* and *Pocillopora acuta* from Kane'ohe Bay, Hawai'i. *15th International Coral Reef Symposium*. 3-8th July 2022, Bremen, Germany.

Selected Awards and honours

2023 Selected to attend and present at the New Phytologist Next Generation Scientists Conference 2023, National University of Singapore

2021 2019 Dean's Award for Outstanding Higher Degree by Research Theses, University of Queensland

2016 Research Training Program (RTP) scholarship, University of Queensland

2012 Dean's Commendation for Academic Excellence (achieved a GPA over 6.6), University of Queensland

2011 Merit Scholarship for academic achievement, University of Queensland

Select Publications

2026 Van Etten J., Han S., Burns J. A., Lhee D., Willoughby A. C., **Stephens T. G.**, Chille E., Sleith R. S., Bhattacharya D., and Yoon H. S. Crawling under the radar: Two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid-containing amoeba lineage. *Journal of Phycology*, 2026. Featured by The New York Times [URL]

2026 **Stephens T. G.**, Kulczyk A. W, and Bhattacharya D. Cosmopolitan gene families with known functions are hotspots for the evolution of novel genes in stony corals. *GBE*, 18:evag072-4, 2026. [URL]

2025 Nandi S., **Stephens T. G.**, Walsh K., Garcia R., Villalpando M. F., Sellares-Blasco R. I., Zubillaga A. L., Croquer A., Bhattacharya D. Shifts in the microbiome and virome are associated with stony coral tissue loss disease (SCTLD). *ISME Communications*, ycaf226, 2025. [Preprint] Featured by the Rutgers Climate and Energy Institute [URL]

2024 **Stephens T. G.**, Van Etten J., McDermott T., Christian W., Chaverra M., Gurney J., Lee Y., Kim H., Hyun Cho C., Chovancek E., Westhoff P., Otte A., Northen T. R., Bowen B. P., Louie K. B., Barry K., Grigoriev I. V., Mock T., Liu S., Miyagishima S., Yoshinaga M, Weber A. P.M., Yoon H. S., and Bhattacharya D. Temporal dynamics in a red alga dominated geothermal feature in Yellowstone National Park. *ISME Communications*, 4:01, 2024. [Preprint] [URL]

2023 **Stephens T. G.**, Strand E. L., Putnam H. M., and Bhattacharya D. Ploidy variation and its implications for reproduction and population dynamics in two sympatric Hawaiian coral species. *Genome Biology and Evolution*, evad149, 2023. [Preprint] [URL]

2022 *Calatreva V., ***Stephens T. G.**, Gabr A., Grossman A. R., and Bhattacharya D. Retrotransposition facilitated the establishment of a primary plastid in the thecate amoeba *Paulinella*. *PNAS*, 119:e2121241119, 2022. *Co-first authorship [URL]

2021 Williams A., Pathmanathan J. S., **Stephens T. G.**, Su X., Chiles E. N., Conetta D., Putnam H. M., and Bhattacharya D. Multi-omic characterization of the thermal stress phenome in the stony coral *Montipora capitata*. *PeerJ*, 9:e12335, 2021. [Preprint] Featured by PeerJ Expert Curations: Molecular Ecology: PeerJ Expert Curations, PeerJ Blog [URL]

2020 **Stephens T. G.**, Gonzalez-Pech R. A., Cheng Y., Mohamed A. R., Burt D. W., Bhattacharya D., Ragan M. A., and Chan C. X. Genomes of the dinoflagellate *Polarella glacialis* encode tandemly repeated single-exon genes with adaptive functions. *BMC Biology*, 18:56, 2020. [Preprint] Featured by IMB News [URL]

2018 **Stephens T. G.**, Ragan M. A., Bhattacharya D., and Chan C. X. Core genes in diverse dinoflagellate lineages include a wealth of conserved dark genes with unknown functions. *Scientific Reports*, 8:17175, 2018. [URL]

Selected Media coverage

2026 Our paper Crawling under the radar: Two novel *Paulinella* species expand knowledge about the ecology and evolution of a primary plastid?containing amoeba lineage covered by The New York Times

2026 Our Sargassum Biofuel project was featured by Rutgers News

2026 Our paper Gene transfer drives community cooperation in geothermal habitats was featured in an article by Phys.org

2026 Our paper Shifts in the microbiome and virome are associated with stony coral tissue loss disease (SCTLD) was covered by Rutgers Climate and Energy Institute

2025 Our paper Metaproteome Analysis of Short-Term Thermal Stress in Three Sympatric Coral Species Reveals Divergent Host Responses was covered by Rutgers Newsroom

2024 Our Sargassum Biofuel project was featured by Rutgers News

2024 Our paper Viruses associated with extremophilic red algal mats reveal signatures of early thermal adaptation was featured by Science

2022 Article by Rutgers Research on our short film The Coral Holobiont Response to Climate Change which won Best Trailer in the Kiez Berlin Film Festival.

2020 Our paper Amoeba Genome Reveals Dominant Host Contribution to Plastid Endosymbiosis covered by Rutgers Today

Selected Service

2023—present Associate Editor: Symbiotic and Parasitic Protists section, Frontiers in Protistology

Patents

2023 Provisional Patent "Protein Markers of Coral Stress" (5431.1025-000:2023-098)

2023 Provisional Patent "Coral Rapid Strip Test" (5431.1024-000:2023-097)